11. The purpose of the study of _____ is to identify the factors of influence and try to control them for better performance.

(a) mean (b) correlation (c) standard deviation (d) skewness

12. The height and weight of a group of persons will have _____ correlation.

(a) positive (b) negative

(c) zero (d) both positive and negative

13. _____ correlation studies the association of two variables with ordinal scale.

(a) A.M. Tuttle rank (b) Croxton and Cowdon rank

(c) Karl Pearson's rank (d) Spearman's rank.

14. _____ presents a graphic description of quantitative relation between two series of facts.

(a) scatter diagram (b) bar diagram (c) pareto diagram (d) pie diagram

15. _____ measures the degree of relationship between two variables.

(a) standard deviation (b) correlation coefficient

(c) moment (d) median

16. The correlation coefficient of x and y is symmetric. Hence

(a) $r_{xy} = r_{yx}$ (b) $r_{xy} > r_{yx}$ (c) $r_{xy} < r_{yx}$ (d) $r_{xy} \neq r_{yx}$

17. If cov $(x, y) = 0$ then its interpretation is

(a) x and y are positively correlated (b) x and y are negatively correlated

(c) x and y are uncorrelated (d) x and y are independent

18. Rank correlation is useful to study data in _____ scale.

(a) ratio (b) ordinal (c) nominal (d) ratio and nominal

19. If $r = 0$ then cov(x, y) is

(a) 0 (b) +1 (c) -1 (d) α

20. If cov$(x, y) = \sigma_x, \sigma_y$ then

(a) $r = 0$ (b) $r = -1$ (c) $r = +1$ (d) $r = \alpha$

II. Give very short answer to the following questions.

21. What is correlation?

22. Write the definition of correlation by A.M. Tuttle.

23. What are the different types of correlation?

24. What are the types of simple correlation?

25. What do you mean by uncorrelated?

26. What you understand by spurious correlation?

12th_Statistics_EM_Unit_4.indd 124 1/28/2020 6:34:36 PM